

SF1811 Optimization

Lecture 10: Linear least-squares problems and orthogonal projections

Chapter 11

Anders Forsgren

Linear least-squares problems

Main question:

How can we find the best approximate solution when $Ax = b$ is inconsistent?

Today:

- Linear model fitting.
- The least-squares problem.
- The normal equations.
- Orthogonal projection onto $\mathcal{R}(A)$.
- Existence and uniqueness of least-squares solutions.
- The minimum-norm solution.

A model-fitting problem

Suppose that measurements

$$(t_1, s_1), \ldots, (t_m, s_m)$$

are given, and that we approximate an unknown function by

$$g(t) \approx \alpha_1 \varphi_1(t) + \cdots + \alpha_n \varphi_n(t),$$

where the basis functions φ_j are known and the coefficients α_j are unknown.

Ideally, $\sum_{j=1}^{n} \alpha_j \varphi_j(t_i) = s_i$, $i = 1, \ldots, m$.

In practice, exact equality may be impossible because

- there may be more measurements than coefficients;
- the model may not be exact;
- the measurements may contain noise.

We seek coefficients that make the discrepancies as small as possible.

From measurements to $Ax \approx b$

Define

$$A = \begin{pmatrix} \varphi_1(t_1) & \cdots & \varphi_n(t_1) \\ \vdots & & \vdots \\ \varphi_1(t_m) & \cdots & \varphi_n(t_m) \end{pmatrix}, \qquad x = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}, \qquad b = \begin{pmatrix} s_1 \\ \vdots \\ s_m \end{pmatrix}.$$

The ideal interpolation conditions become

$$Ax = b.$$

If this system is inconsistent, we minimize the sum of squared residuals:

$$\sum_{i=1}^{m} \left(\sum_{j=1}^{n} \alpha_j \varphi_j(t_i) - s_i \right)^2 = \|Ax - b\|^2.$$

Model fitting leads naturally to a linear least-squares problem.

The linear least-squares problem

The linear least-squares problem is

$$(LS) \qquad \underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2}\|Ax - b\|^2.$$

The vector

$$r(x) = b - Ax$$

is called the residual.

Thus (LS) seeks a vector x for which the residual norm

$$\|r(x)\| = \|b - Ax\|$$

is as small as possible.

The factor $1/2$ does not affect the minimizers, but simplifies the derivatives.

Least squares as quadratic optimization

The objective function is

$$\begin{aligned} f(x) &= \frac{1}{2}(Ax - b)^{\mathsf{T}}(Ax - b) \\ &= \frac{1}{2}x^{\mathsf{T}}A^{\mathsf{T}}Ax - b^{\mathsf{T}}Ax + \frac{1}{2}b^{\mathsf{T}}b. \end{aligned}$$

Hence f is quadratic, with

$$H = A^{\mathsf{T}}A, \qquad c = -A^{\mathsf{T}}b.$$

The gradient and Hessian are

$$\nabla f(x) = (A^{\mathsf{T}}Ax - A^{\mathsf{T}}b)^{\mathsf{T}} = (Ax - b)^{\mathsf{T}}A,$$

and

$$F(x) = A^{\mathsf{T}}A.$$

For every $d \in \mathbb{R}^n$, $d^{\mathsf{T}}A^{\mathsf{T}}Ad = \|Ad\|^2 \geq 0$. Therefore, $A^{\mathsf{T}}A \succeq 0$, and every linear least-squares problem is convex.

The normal equations

Since the least-squares problem is unconstrained and convex, $\hat{x}$ is a minimizer if and only if

$$\nabla f(\hat{x}) = 0.$$

Thus

$$A^{\mathsf{T}} A\hat{x} - A^{\mathsf{T}} b = 0,$$

or equivalently,

$$\boxed{A^{\mathsf{T}} A\hat{x} = A^{\mathsf{T}} b.}$$

These equations are called the normal equations.

Equivalently,

$$A^{\mathsf{T}}(b - A\hat{x}) = 0.$$

Solving a least-squares problem means solving the normal equations.

Geometric meaning of the normal equations

The vector $A\hat{x}$ belongs to $\mathcal{R}(A)$.

The normal equations give

$$A^{\mathsf{T}}(b - A\hat{x}) = 0,$$

which means

$$b - A\hat{x} \in \mathcal{N}(A^{\mathsf{T}}).$$

Using

$$\mathcal{N}(A^{\mathsf{T}}) = \mathcal{R}(A)^{\perp},$$

we obtain

$$\boxed{b - A\hat{x} \perp \mathcal{R}(A).}$$

Thus $A\hat{x}$ is the point in $\mathcal{R}(A)$ closest to b.

Orthogonal projection onto $\mathcal{R}(A)$

The least-squares problem can equivalently be written as

$$\begin{array}{ll} \text{minimize}_y & \frac{1}{2}\|y - b\|^2 \\ \text{subject to} & y \in \mathcal{R}(A). \end{array}$$

The optimal point

$$\hat{y} = A\hat{x}$$

is characterized by

$$\hat{y} \in \mathcal{R}(A), \qquad b - \hat{y} \in \mathcal{R}(A)^{\perp}.$$

Therefore,

$$\hat{y} = \text{proj}_{\mathcal{R}(A)}\, b.$$

The projected point $\hat{y} = A\hat{x}$ is unique, even if the coefficient vector $\hat{x}$ is not unique.

Least squares is an orthogonal projection problem.

Why does a least-squares solution always exist?

The normal equations are

$$A^{\mathsf{T}}Ax = A^{\mathsf{T}}b.$$

They have a solution because

$$\mathcal{R}(A^{\mathsf{T}}A) = \mathcal{R}(A^{\mathsf{T}}).$$

Indeed,

$$\mathcal{N}(A^{\mathsf{T}}A) = \mathcal{N}(A),$$

since

$$A^{\mathsf{T}}Ax = 0 \quad \Longleftrightarrow \quad x^{\mathsf{T}}A^{\mathsf{T}}Ax = \|Ax\|^2 = 0 \quad \Longleftrightarrow \quad Ax = 0.$$

Taking orthogonal complements gives $\mathcal{R}(A^{\mathsf{T}}A) = \mathcal{R}(A^{\mathsf{T}})$.

Since $A^{\mathsf{T}}b \in \mathcal{R}(A^{\mathsf{T}})$, the normal equations are consistent.

Proposition

Every linear least-squares problem has at least one minimizer.

Full column rank

Suppose that the columns of A are linearly independent. Then

$$\mathcal{N}(A) = \{0\}.$$

For every $d \neq 0$, $d^T A^T A d = \|Ad\|^2 > 0$. Hence $A^T A \succ 0$.
Therefore, the normal equations have a unique solution:

$$\hat{x} = (A^T A)^{-1} A^T b.$$

In computation, the inverse is not formed explicitly. Instead, solve

$$A^T A \hat{x} = A^T b,$$

for example using a factorization of $A^T A$.

Linearly independent columns give a unique least-squares solution.

Linearly dependent columns

Suppose that the columns of A are linearly dependent. Then

$$\mathcal{N}(A) \neq \{0\}$$

and $A^{\mathsf{T}}A$ is singular.

If $\hat{x}$ solves the normal equations and $z \in \mathcal{N}(A)$, then

$$A(\hat{x} + z) = A\hat{x}.$$

Thus $\hat{x} + z$ has the same residual and is also a least-squares solution.

Conversely, if x_1 and x_2 are least-squares solutions, then

$$Ax_1 = Ax_2,$$

so $x_1 - x_2 \in \mathcal{N}(A)$.

Hence all least-squares solutions have the form

$$\boxed{x = \hat{x} + z, \qquad z \in \mathcal{N}(A).}$$

A unique fitted vector, but nonunique coefficients

Even when the coefficient vector is not unique, the fitted vector is unique.

Let x_1 and x_2 solve the normal equations. Then

$$A^{\mathsf{T}}A(x_1 - x_2) = 0.$$

Therefore,

$$\begin{aligned} \|A(x_1 - x_2)\|^2 &= (x_1 - x_2)^{\mathsf{T}}A^{\mathsf{T}}A(x_1 - x_2) \\ &= 0. \end{aligned}$$

Hence

$$Ax_1 = Ax_2.$$

Thus:

- the fitted vector $A\hat{x}$ is unique;
- the residual $b - A\hat{x}$ is unique;
- the coefficients $\hat{x}$ need not be unique.

Different coefficient vectors may describe exactly the same fit.

The minimum-norm least-squares solution

When the least-squares solution is not unique, select the solution with smallest Euclidean norm:

$$\hat{x}_{\mathrm{MN}} = \operatorname{argmin} \{\|x\| : x \text{ solves the least-squares problem}\}.$$

Using the orthogonal decomposition

$$\mathbb{R}^n = \mathcal{R}(A^{\mathsf{T}}) \oplus \mathcal{N}(A),$$

write a least-squares solution as

$$x = x_R + x_N, \qquad x_R \in \mathcal{R}(A^{\mathsf{T}}), \qquad x_N \in \mathcal{N}(A).$$

Since $Ax = Ax_R$, the component x_N does not affect the fitted vector.

Therefore, the minimum-norm solution is characterized by

$$\boxed{\hat{x}_{\mathrm{MN}} \in \mathcal{R}(A^{\mathsf{T}}).}$$

Why is the minimum-norm solution unique?

Every least-squares solution can be written as

$$x = \hat{x}_{\mathrm{MN}} + z, \qquad z \in \mathcal{N}(A).$$

Since

$$\hat{x}_{\mathrm{MN}} \in \mathcal{R}(A^{\mathsf{T}}) = \mathcal{N}(A)^{\perp},$$

we have

$$\hat{x}_{\mathrm{MN}} \perp z.$$

The Pythagorean theorem gives

$$\|x\|^2 = \|\hat{x}_{\mathrm{MN}} + z\|^2 = \|\hat{x}_{\mathrm{MN}}\|^2 + \|z\|^2.$$

Therefore,

$$\|x\| \geq \|\hat{x}_{\mathrm{MN}}\|,$$

with equality only when $z = 0$.

Removing the nullspace component gives the unique shortest solution.

The pseudoinverse

There is a unique matrix A^+ such that the minimum-norm least-squares solution is

$$\hat{x}_{\mathrm{MN}} = A^+ b.$$

The matrix A^+ is called the Moore–Penrose pseudoinverse of A.

If A has full column rank, then

$$A^+ = (A^{\mathsf{T}} A)^{-1} A^{\mathsf{T}}.$$

For a rank-deficient matrix, the pseudoinverse can be constructed using a singular value decomposition.

The pseudoinverse selects the minimum-norm solution automatically.

A small least-squares example

Consider

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

The normal equations are

$$\begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Solving gives $\hat{x} = \begin{pmatrix} \frac{7}{6} \\ \frac{1}{2} \end{pmatrix}$. Thus $A\hat{x} = \begin{pmatrix} \frac{7}{6} \\ \frac{5}{3} \\ \frac{13}{6} \end{pmatrix}$, $b - A\hat{x} = \begin{pmatrix} -\frac{1}{6} \\ \frac{1}{3} \\ -\frac{1}{6} \end{pmatrix}$.

The example as line fitting

The rows of A correspond to the measurement locations

$$t_1 = 0, \qquad t_2 = 1, \qquad t_3 = 2,$$

and the model

$$g(t) = x_1 + x_2 t.$$

The observations are

$$(0, 1), \qquad (1, 2), \qquad (2, 2).$$

The least-squares line is

$$\hat{g}(t) = \frac{7}{6} + \frac{1}{2} t.$$

Least squares balances all residuals rather than fitting every measurement exactly.

Main ideas

After this lecture, you should be able to:

- formulate a linear model-fitting problem as a least-squares problem;
- derive and solve the normal equations;
- interpret least squares as orthogonal projection onto $\mathcal{R}(A)$;
- explain why a least-squares solution always exists;
- characterize uniqueness using the columns of A;
- distinguish between the unique fitted vector and possibly nonunique coefficients;
- characterize the minimum-norm solution by $\hat{x}_{\mathrm{MN}} \in \mathcal{R}(A^{\mathsf{T}})$;
- explain the role of the pseudoinverse.

The normal equations say that the residual is orthogonal to every model direction.